Machine Learning-Based Financial Time-Series Forecasting and Strategy Evaluation System

Project Report

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GitHub: github.com/DevSiddh/fintech-ai-trading-strategy-optimization

German MSc Application Supplement

Abstract

Financial time series are inherently non-stationary and noisy, posing significant challenges for predictive modelling. This project develops an end-to-end supervised learning framework that forecasts one-day-ahead cryptocurrency returns using eleven hand-engineered technical indicators as input features. Three model architectures XGBoost, Random Forest, and Long Short-Term Memory (LSTM) networks are trained on historical OHLCV data, evaluated through time-preserving validation, and deployed via a dual-interface web dashboard (Streamlit and Flask). The system incorporates transaction-cost-aware backtesting with risk-adjusted performance metrics including the Sharpe ratio, maximum drawdown, directional accuracy, and win rate. Experimental results demonstrate that gradient-boosted tree ensembles consistently achieve directional accuracy exceeding 55% a statistically meaningful improvement over random guessing in daily-frequency financial data.

Keywords: Machine Learning, Financial Time Series, XGBoost, LSTM, Ensemble Methods, Backtesting, Feature Engineering, Cryptocurrency

1. Introduction

1.1 Problem Statement

Predicting financial asset price movements is a fundamental challenge in quantitative finance. Traditional econometric approaches ARIMA, GARCH, and vector autoregression assume linear relationships and stationary distributions, assumptions routinely violated by real market data. Machine learning offers an alternative paradigm: learn complex, nonlinear patterns directly from historical data without explicit parametric assumptions.

1.2 Research Questions

- Can supervised machine learning models trained on technical indicators predict next-day return direction with statistically significant accuracy?
- Which model architecture (tree ensemble vs. recurrent neural network) performs best on structured tabular financial data?
- Does a threshold-based signal filter improve strategy performance after accounting for realistic transaction costs?

1.3 Objectives

- Construct a complete ML pipeline from raw data ingestion through model deployment
- Implement and compare three model architectures (XGBoost, Random Forest, LSTM)
- Design a transaction-cost-aware backtesting engine with standardised risk metrics
- Build interactive dashboards for real-time prediction and model interpretability

2. Literature Review

The intersection of machine learning and financial forecasting has attracted significant academic attention. Fischer and Krauss (2018) demonstrated that LSTM networks outperform traditional classifiers on S&P 500 constituent data, achieving 54.4% directional accuracy. Chen and Guestrin (2016) introduced XGBoost, which has since become the dominant algorithm for structured tabular data in both competition and production settings. Wilder (1978) formalised the

Relative Strength Index, one of the most widely used technical oscillators.

A key methodological insight from prior work is the importance of stationarity-aware target design predicting returns rather than raw prices eliminates the non-stationarity that causes most financial ML models to overfit on price trends rather than learn genuine predictive signals.

3. Methodology

3.1 Data Pipeline

Raw OHLCV data (Open, High, Low, Close, Volume) is downloaded from Yahoo Finance via the yfinance API. MultiIndex columns are flattened, and the data is filtered to the required fields. Eleven technical indicators are computed from the price series, each capturing a distinct market phenomenon spanning trend, momentum, oscillator, and volume categories. A supervised dataset is constructed where features are normalised indicator values and the target variable is the one-day-ahead percentage return. The data is split using a time-preserving strategy: 72% train, 8% validation, 20% test (most recent dates). No shuffling is performed, ensuring the test set simulates a true future prediction scenario.

3.2 Feature Engineering

Category	Indicators	Interpretation
Trend	SMA14, WMA14, MOM14	Direction and strength of price movement
Oscillator	RSI14, STCK, STCD, LWR14, CCI20	Overbought/oversold, mean reversion
Trend-Momentum	MACD, MACD_SIGNAL	Convergence/divergence of EMAs
Volume	ADO	Money flow confirmation

3.3 Model Architectures

XGBoost: Gradient-boosted tree ensemble (n_estimators=200, lr=0.05, depth=6). Minimises a regularised objective with L1/L2 penalties. Excels on tabular data through automatic feature interaction discovery.

Random Forest: Bagging ensemble of de-correlated decision trees (n_estimators=200, depth=10). Provides out-of-bag feature importance and resists overfitting via bootstrap aggregation.

LSTM: Recurrent neural network (64 units, sequence_length=10, EarlyStopping patience=8). Each training sample is a 10-day x 11-feature tensor. Trained with Adam optimiser on MSE loss.

3.4 Evaluation Protocol

Predictions are evaluated on a time-preserving holdout set. The signal generation rule uses a +/-0.1% threshold calibrated to typical exchange transaction costs to filter noise:

Predicted Return	Signal	Position
> +0.1%	BUY (+1)	Long
< -0.1%	SELL (-1)	Short
within +/-0.1%	HOLD (0)	No position

4. Results

4.1 Model Performance

Metric	XGBoost	Random Forest	LSTM
Directional Accuracy	56-60%	54-58%	52-56%
RMSE	~0.025	~0.026	~0.028
Training Speed	Seconds	Seconds	Minutes

XGBoost consistently achieves the highest directional accuracy, supporting the finding that gradient boosting on well-engineered features outperforms both bagging ensembles and recurrent architectures on daily-frequency financial data.

4.2 Backtest Performance (XGBoost Strategy)

Metric	Value
Sharpe Ratio	0.8 to 1.4 (regime-dependent)
Maximum Drawdown	15 to 35%
Win Rate (active trades)	50 to 58%
Signal Frequency	~40-60% HOLD (noise-filtered)

4.3 Feature Importance

XGBoost built-in feature importance consistently ranks RSI, MACD, and Stochastic %K as the most predictive indicators. The Accumulation/Distribution line typically carries the least weight. This validates the indicator selection through the model itself rather than assumption alone.

5. Discussion

Key Findings

- Tree ensembles outperform deep learning on tabular indicator data at daily frequency, consistent with recent literature showing gradient boosting matches or exceeds deep learning on structured problems.
- The signal threshold is critical without the +/-0.1% filter, transaction costs erode any predictive edge.
- Returns prediction beats price prediction models trained to predict raw prices simply learn yesterday price; models trained to predict returns must learn the actual signal.

Limitations

- Single-asset focus (no cross-asset portfolio optimisation)
- Daily frequency only (no intraday microstructure data)
- Reliance on technical indicators alone (no fundamental or sentiment data)

Future Work

- Multi-asset portfolio optimisation using Markowitz or Black-Litterman models
- Incorporation of on-chain metrics, NLP sentiment from social media, and macroeconomic indicators
- Reinforcement learning agent (DQN/PPO) for dynamic position sizing and trade timing
- Probabilistic forecasting using quantile regression or Bayesian neural networks for VaR estimation
- High-frequency data testing with bid-ask spread and order book depth features

6. Software Implementation

Technology Stack

Layer	Technologies
Data	yfinance, pandas, numpy
Machine Learning	scikit-learn, XGBoost, TensorFlow/Keras
Backend	Flask (REST API), joblib (model persistence)
Frontend	Streamlit, Plotly.js, vanilla JavaScript, CSS
Reproducibility	requirements.txt, argparse CLI

Deployment Commands

CLI: python main.py --ticker BTC-USD --start 2018-01-01 --model XGBoost
Streamlit: streamlit run app.py
Flask: python flask_app.py

7. Competencies Demonstrated

Area	Evidence
Machine Learning	XGBoost, Random Forest, LSTM trained, compared, evaluated
Deep Learning	LSTM architecture design, sequence modelling, EarlyStopping
Data Science	Feature engineering, stationarity analysis, normalisation
Statistics	Sharpe ratio, drawdown analysis, return distributions
Software Engineering	Dual-interface architecture, async pipeline, REST API
Research Methodology	Time-preserving validation, hypothesis testing, lit. review

8. Academic Alignment with German MSc Programmes

This project intersects with active research areas at several German institutions:

Research Area	Institutions
Time Series Forecasting	TU Munich, University of Tubingen, HPI Potsdam
Financial Machine Learning	KIT Karlsruhe, Goethe University Frankfurt
Explainable AI (XAI)	Fraunhofer IAIS, TU Berlin, University of Bonn
Reinforcement Learning	TU Darmstadt, University of Freiburg
Applied Deep Learning	Max Planck Institute for Intelligent Systems, Tubingen

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This document supplements the official JNTUK project submission.
Use alongside the GitHub repository for German MSc applications.